MODULE 41 · EXAM STRATEGY - ALL SECTIONS

Confusion Pairs: The One Word That Separates Them

Eighteen pairs and trios the exam sets against each other, from Bigtable and BigQuery to system lag and data freshness, each with the word that decides it.

SECTIONS All five MODULE 41 of 45 SCENARIOS 8

admin@sthithapragna.com

Most wrong answers are a good product in the wrong place. **Find the word in the question that rules one out.**

admin@sthithapragna.com

Bigtable or BigQuery?

BIGTABLE

Serve by key, fast

- Single-digit millisecond reads by row key
- Huge write throughput, time series
- No SQL joins across tables
- Sized in nodes

BIGQUERY

Analyse with SQL

- Scans and aggregates billions of rows
- Ad hoc SQL, joins, BI tools
- Seconds, not milliseconds
- Serverless, slots

THE WORD

Milliseconds and a key: Bigtable. Analysts and SQL: BigQuery.

Spanner, Cloud SQL or AlloyDB?

PICK	WHEN THE QUESTION SAYS
Cloud SQL	Lift a MySQL, PostgreSQL or SQL Server app. One region, up to 64 TB.
AlloyDB	PostgreSQL, and analytics on live transactions, via the columnar engine.
Spanner	Global or multi-region writes, horizontal scale, 99.999%.
Firestore	Documents for mobile and web apps, with offline sync.

THE WORD

Global and consistent: Spanner. PostgreSQL plus analytics: AlloyDB. Familiar and modest: Cloud SQL.

admin@sthithapragna.com

Dataflow, Spark, Data Fusion or Dataform?

PICK	WHEN THE QUESTION SAYS
Dataflow	New pipelines, streaming and batch in one model, serverless.
Managed Spark	Existing Spark or Hadoop jobs, moved with few changes.
Data Fusion	Visual, drag-and-drop pipelines, many connectors, little code.
Dataform	SQL transformations inside BigQuery, in Git, with tests.

THE WORD

Existing Spark: Managed Spark. Streaming from scratch: Dataflow. Already in BigQuery: Dataform.

admin@sthithapragna.com

Airflow, Workflows or Cloud Scheduler?

PICK	WHEN THE QUESTION SAYS
Managed Airflow	Many data tasks, dependencies, retries, backfills, Python DAGs.
Workflows	A short sequence of API or HTTP calls, serverless, YAML.
Cloud Scheduler	One job, on a timer. A cron line, nothing more.

THE WORD

Backfill or a DAG: Airflow. Chained API calls: Workflows. Just a timer: Scheduler.

admin@sthithapragna.com

Pub/Sub or Managed Kafka?

PUB/SUB

Google-native, serverless

- No partitions or brokers to size
- Global, scales automatically
- Push, pull and export subscriptions
- At-least-once, seek to replay

MANAGED KAFKA

Kafka, run for you

- Existing Kafka clients unchanged
- Partitions and offsets you control
- Kafka Connect included
- Sized in vCPU and memory

THE WORD

Kafka clients or Kafka skills already in place: Managed Kafka. Otherwise, Pub/Sub.

admin@sthithapragna.com

Five ways in, five different jobs

PICK	WHEN THE QUESTION SAYS
Database Migration Service	Move a database to Cloud SQL or AlloyDB, then cut over.
Datastream	Keep streaming changes from a database into BigQuery.
BigQuery Data Transfer Service	Scheduled loads from SaaS apps or other warehouses.
Storage Transfer Service	Files from other clouds or on-premises, over the network.
Transfer Appliance	Hundreds of terabytes, and the network would take months.

THE WORD

Migrate and cut over: DMS. Keep replicating for analytics: Datastream.

Who holds the key?

OPTION	WHO CONTROLS THE KEY
Default encryption	Google creates and rotates it. Nothing to do.
CMEK	You, in Cloud KMS. Rotate, disable, audit. HSM optional.
CSEK	You send it with each request. Storage and Compute only.
Cloud EKM	An external key manager. Google never holds the material.

THE WORD

Key outside Google entirely: EKM. Control and audit inside Google: CMEK. BigQuery never takes CSEK.

admin@sthithapragna.com

Two kinds of masking

SENSITIVE DATA PROTECTION

Change the data

- Inspect and classify, then transform
- Redact, tokenise, or shift dates
- The stored copy is de-identified
- Before data lands, or for extracts

BIGQUERY DATA MASKING

Change what each reader sees

- Data policies on policy tags
- Raw for some readers, masked for others
- Stored data unchanged
- Hash, nullify, last four, and more

THE WORD

Nobody may ever see raw values: de-identify with SDP. Some may, some may not: dynamic masking.

Materialized view, BI Engine or the cache?

PICK	HELPS WHEN
Materialized view	The same aggregation over a big, changing table. Stays fresh.
BI Engine	Interactive dashboards, sub-second, on modest tables, in memory.
Results cache	The identical query again, on unchanged data. Free, about 24 hours.

THE TIEBREAKER

Data keeps arriving: the cache misses, the materialized view does not.

admin@sthithapragna.com

Partitioning or clustering?

PARTITIONING

Coarse slices

- One column: date, timestamp or integer
- Prunes whole partitions, cost known up front
- Expiry and filters per partition
- Up to 10,000 partitions

CLUSTERING

Fine sorting

- Up to four columns, order matters
- Prunes blocks within each partition
- Good for high-cardinality filters
- Cost known only after running

THE PATTERN

Filter on date and on customer? Partition by date, cluster by customer. Usually both.

admin@sthithapragna.com

Update, drain or cancel?

ACTION	IN-FLIGHT DATA	REACH FOR IT WHEN
Update	Kept, with state	New code, compatible graph
Drain	Finished, windows closed early	Stopping, or an incompatible change
Cancel	Buffered data may be lost	Stopping now matters more than data

THE WORD

Keep state: update. Lose nothing but restart: drain. Neither matters: cancel.

admin@sthithapragna.com

Tumbling, hopping or session windows?

WINDOW	SHAPE	EXAMPLE QUESTION
Tumbling, fixed	Back to back, no overlap	Total kWh per half hour
Hopping, sliding	Overlapping, with a period	Five-minute average, every minute
Session	Per key, closed by a gap	One app visit, ended by 30 minutes idle

THE WORD

Per interval, rolling, or idle: those three words pick the window.

admin@sthithapragna.com

Time travel, snapshot or replica?

PICK	PROTECTS AGAINST	LASTS
Time travel	A recent mistake: bad update, dropped table	2 to 7 days, automatic
Table snapshot	Needing a point-in-time copy for longer	Until it expires
Replica or managed DR	Losing the region	Continuous

THE WORD

Mistake this week: time travel. Audit copy for a year: snapshot. Region lost: managed DR.

Authorised view or sharing listing?

AUTHORISED VIEW

Inside your organisation

- A view in another dataset, authorised on the source
- Readers need no access to the base tables
- Filter rows and columns in SQL
- Shared by IAM on the view's dataset

BIGQUERY SHARING

Across organisations

- Publish a listing in an exchange
- Subscribers get a linked, read-only dataset
- No copies, always current
- Subscribers pay for their own queries

THE WORD

Another team: authorised view. Another company, at scale: a sharing listing.

admin@sthithapragna.com

VPC Service Controls or IAM?

IAM

Who may do what

- Grants to identities
- Least privilege, roles, conditions
- A stolen key still works
- Answers: may this person read it?

VPC SERVICE CONTROLS

Where data may go

- A perimeter around projects
- Blocks copying data outside it
- Stops a stolen key working from outside
- Answers: may it leave this boundary?

THE WORD

Exfiltration, even by authorised users: a perimeter. Who can access: IAM. Most designs need both.

admin@sthithapragna.com

System lag or data freshness?

SYSTEM LAG

Is a step stuck?

- Longest time an item has waited or processed
- Measured in processing time
- Rises: slow code, a hot key, an external call
- Points to the pipeline

DATA FRESHNESS

Are results stale?

- How far output trails event time
- Includes late data from sources
- Rises: input late, or pipeline behind
- Points to what users see

THE PATTERN

Alert users on freshness. Debug the pipeline with lag.

> The customer app shows each meter's latest reading. It needs answers in about ten milliseconds, at tens of thousands of reads a second.
>
> Every reading is already loaded into BigQuery for analysts.

Where should the app read from?

A The BigQuery readings table, accelerated by a BI Engine reservation

B The BigQuery readings table, clustered on meter_id for pruning

C A Bigtable table keyed on meter ID, written by the pipeline

D Cloud SQL read replicas, loaded each hour from BigQuery

◆ PAUSE AND DECIDE

Where should the app read from?

A BI Engine

For dashboards, not ten-millisecond lookups.

B Clustered BigQuery

Faster scans, still seconds, not milliseconds.

C Bigtable by key

Single-digit milliseconds at huge read rates.

D Cloud SQL replicas

An hour stale, and a strain at that rate.

THE SIGNAL

Milliseconds by key at high volume: Bigtable, not BigQuery.

> The tariff service runs on PostgreSQL in London. The pricing team wants heavy analytical queries on live transactional data, with no ETL.
>
> The service must stay PostgreSQL-compatible, in one region.

Which database fits best?

A AlloyDB for PostgreSQL, using its columnar engine

B Spanner, in a multi-region configuration for scale

C Cloud SQL, with hourly exports into BigQuery for analysis

D Bigtable, with the transactions stored as wide rows

◆ PAUSE AND DECIDE

Which database fits best?

A **AlloyDB**
PostgreSQL, with analytics on live data.

B **Spanner**
Global scale nobody asked for.

C **Cloud SQL plus exports**
That is ETL, and an hour stale.

D **Bigtable**
Not relational, not PostgreSQL.

THE SIGNAL

PostgreSQL plus analytics on live data: AlloyDB.

> Analysts transform raw tables into marts entirely inside BigQuery, with SQL scripts run by hand.
>
> They want version control, dependency order, and automatic data tests, still in SQL.

What should they adopt?

A Data Fusion pipelines, built in the visual designer

B Dataflow pipelines, written in Java with Apache Beam

C Managed Spark jobs, rewriting the scripts in Spark SQL

D Dataform, with SQLX files in Git and assertions

◆ PAUSE AND DECIDE

What should they adopt?

A **Data Fusion**
Visual, and moves data rather than staying in SQL.

B **Dataflow**
Code, not SQL, and data leaves BigQuery.

C **Managed Spark**
A cluster and a rewrite for SQL that already works.

D **Dataform**
SQL in Git, dependencies, assertions as tests.

THE SIGNAL

SQL already in BigQuery, needing Git and tests: Dataform.

> Once a night, a cache in the tariff service must be refreshed by calling one HTTP endpoint.
>
> Nothing depends on it, and nothing else runs with it. The team wants the simplest managed option.

What should they use?

A. A Managed Airflow DAG with one HTTP operator task

B. A Cloud Scheduler job that calls the endpoint

C. A Workflows definition, started by an Eventarc trigger

D. A Dataflow batch job that makes the HTTP request

◆ PAUSE AND DECIDE

What should they use?

A Airflow DAG

An entire environment for one call.

B Cloud Scheduler

One call, on a timer. Nothing more needed.

C Workflows

Built for chains of calls, and still needs a timer.

D Dataflow

A data pipeline for no data at all.

THE SIGNAL

One call on a timer: Cloud Scheduler.

> An on-premises Oracle billing database must stay in service. Analysts want its changes in BigQuery within minutes, continuously.
>
> Nobody plans to move the application itself.

Which service should carry the changes?

A Database Migration Service, then a cutover to AlloyDB

B Datastream, streaming change data into BigQuery

C BigQuery Data Transfer Service, on a nightly schedule

D Storage Transfer Service, copying nightly export files

◆ PAUSE AND DECIDE

Which service should carry the changes?

A DMS

For moving the database, which nobody wants.

B Datastream

Continuous change capture into BigQuery.

C DTS

Scheduled, not continuous, and not for Oracle CDC.

D Storage Transfer

Moves files, a day at a time.

THE SIGNAL

Keep replicating changes for analytics: Datastream.

> A regulator requires that encryption keys for German customer data in BigQuery are never held by Google.
>
> The keys must live in a third-party key manager that Larkspur controls, and every use must be justified.

Which option meets the rule?

A Google default encryption, with keys rotated automatically

B CMEK in Cloud KMS, using the HSM protection level

C CSEK sent with every query by the client

D Cloud EKM, with keys in a partner external key manager

◆ PAUSE AND DECIDE

Which option meets the rule?

A Default

Google holds and manages the keys.

B CMEK with HSM

Your key, but held in Google's HSMs.

C CSEK

Not supported by BigQuery.

D Cloud EKM

Key material never leaves the external manager.

THE SIGNAL

Keys never held by Google: Cloud EKM.

> Dozens of reports compute the same daily totals from a four-terabyte readings table. New rows arrive every few minutes.
>
> The reports must include the newest rows, and the team wants each report to scan far less data.

What should the team build?

A A materialized view holding the daily totals

B Nothing, and rely on the query results cache

C A BI Engine reservation in front of the table

D A scheduled query rebuilding a totals table each night

◆ PAUSE AND DECIDE

admin@sthithapragna.com

What should the team build?

A Materialized view
Precomputed, refreshed, and current.

B Results cache
Misses, because the table keeps changing.

C BI Engine
Speeds dashboards, but the table is huge.

D Nightly totals
Small scans, but up to a day stale.

THE SIGNAL

The same aggregate over a changing table: a materialized view.

> Each month end, auditors want a frozen copy of the billing tables as they stood, kept for thirteen months.
>
> The tables change only a little each month, and storage cost should stay low.

What should the team use?

A Time travel, with the window set to its maximum

B A dataset replica in a second region

C A snapshot each month, expiring after 13 months

D A full table copy each month into an archive dataset

◆ PAUSE AND DECIDE

What should the team use?

A Time travel

Seven days at most, not thirteen months.

B Replica

Always current, never frozen.

C Monthly snapshots

Frozen, read-only, billed only for changes.

D Full copies

Frozen, but pays for every byte again.

THE SIGNAL

A frozen copy for months, cheaply: table snapshots with expiry.

Module 41 in five lines

- Milliseconds by key: **Bigtable**. SQL analysis: **BigQuery**. Live analytics in PostgreSQL: **AlloyDB**.
- Existing Spark: **Managed Spark**. SQL in BigQuery: **Dataform**. One timer: **Scheduler**.
- Migrate: **DMS**. Keep replicating: **Datastream**. Key outside Google: **EKM**.
- Changing table, repeated aggregate: **materialized view**. Frozen copy: **snapshot**.
- Find the **one word** that rules a product out, then choose.

> COMING UP
>
> Next: module 42, how PDE questions are written.